Spec → Code: RID010

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1. Spec

REPORT ID

RID010

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report lists subjects with gaps or overlaps in Toxicity Grade Dates, with AE Onset Date not equal to the earliest Toxicity Grade Onset Date OR with AE End Date that is not equal to the latest Toxicity Grade Stop Date.

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. AE Seq
5. AE Term
6. AE Onset Date
7. AE End Date
8. Tox Grade Onset Date
9. Tox Grade Stop Date
10. Error Message

LOGIC

This report lists subjects with the following conditions:

1) Add entry Toxicity Grade Date overlaps. Output an error message that says "Overlap with Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy"

2) Add entry Toxicity Grade Date gaps. Output an error message that says "Gap with previous Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy"

3) AE Onset Date does not equal the minimum date of Toxicity Grade Onset Date in the add entry section of the same AE. Output an error message that says "AE Onset not earliest Tox Grade Onset"

4a) AE End Date does not equal the maximum date of Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

4b) AE End Date is ongoing but there is no blank Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

4c) AE End Date has a date but there is a blank Toxicity Grade Stop Date in the add entry section of the same AE. Output an error message that says "AE End not latest Tox Grade Stop"

5) AE Term is completed but Add-entry section is blank, either because it's not initialized or because all the records are completed with UNK/ND/NA. Output an error message that says "Add entry missing"

Report should not display deleted records.

Add-entry records with both dates as UNK/ND/NA are not be considered for scenarios 1 to 4 when there are other records with data entered.

If "Randomized only" parameter selection checkbox is checked, the report should only output subjects with Randomization number that is not null.

REPORT SCENARIOS

Test Case Workbook

SORTING

Country, Site, Subject, AE Seq

2. Logic Understanding

This report cross-checks each AE record's Onset/End dates against the date range covered by its nested 'add entry' Toxicity Grade Onset/Stop Date records, flagging gaps, overlaps, mismatched earliest/latest dates, and inconsistent ongoing/blank-stop-date logic, plus flagging AEs entirely missing their add-entry toxicity data. Placeholder UNK/ND/NA-only add-entry records are ignored when real data exists elsewhere, deleted records are excluded, and an optional parameter can restrict output to randomized subjects only. Several nuances -- like how ties in dates are handled or whether 'deleted' applies at add-entry level -- are inferred rather than explicit in the spec.

1. For a given AE record, examine all its 'add entry' Toxicity Grade Onset Date / Toxicity Grade Stop Date pairs and check whether any two add-entry date ranges overlap each other. If so, output the error 'Overlap with Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy' using the specific overlapping dates.

2. Similarly, check whether there is a gap between the Stop Date of one add-entry record and the Onset Date of the next add-entry record (i.e., they are not contiguous) for the same AE. If so, output the error 'Gap with previous Tox Grade Onset dd-mon-yyyy and Tox Stop dd-mon-yyyy' using the specific dates involved.

3. Compare the AE Onset Date to the minimum (earliest) Toxicity Grade Onset Date among all add-entry records for that same AE. If they are not equal, output 'AE Onset not earliest Tox Grade Onset'.

4. Compare the AE End Date to the maximum (latest) Toxicity Grade Stop Date among all add-entry records for that same AE. If they are not equal, output 'AE End not latest Tox Grade Stop' (scenario 4a).

5. As a variant of the AE End check: if AE End Date indicates the AE is ongoing (no end date) but none of the add-entry records for that AE has a blank Toxicity Grade Stop Date, this is inconsistent -- output 'AE End not latest Tox Grade Stop' (scenario 4b).

6. As another variant: if AE End Date has an actual completed date but there exists an add-entry record with a blank Toxicity Grade Stop Date for that same AE, this is inconsistent -- output 'AE End not latest Tox Grade Stop' (scenario 4c).

7. Before applying scenarios 1-4, exclude any add-entry record where BOTH Toxicity Grade Onset Date and Toxicity Grade Stop Date are UNK/ND/NA, but only in cases where other add-entry records for that same AE contain actual entered data -- meaning these placeholder-only records are ignored as noise when real data exists elsewhere.

8. Separately, check whether the AE Term is completed (i.e., the AE record itself is filled in) but the entire add-entry section for that AE is blank -- either because no add-entry records were ever created, or because all existing add-entry records only contain UNK/ND/NA in both date fields. In that case, output 'Add entry missing' (scenario 5) instead of running scenarios 1-4 for that AE.

9. Exclude any deleted records (whether at the AE level or the add-entry level, inferred) from all of the above checks and from the report output entirely. *(inferred)*

10. If the 'Randomized only' parameter checkbox is selected when running the report, filter the final output to only include subjects whose Randomization Number is not null; otherwise include all subjects regardless of randomization status.

11. Present all resulting error rows sorted by Country, Site, Subject, AE Seq, with each row showing Country, Site, Subject Number, AE Seq, AE Term, AE Onset Date, AE End Date, Tox Grade Onset Date, Tox Grade Stop Date, and the specific Error Message generated from steps 1-8.

3. Spec Review

Readiness: Needs revision — The spec has a blocking gap in defining what constitutes an overlap/gap in Toxicity Grade Dates, plus several warnings around UNK/ND/NA handling ambiguity, missing/underspecified Report Scenarios, and column population rules for multi-record cases -- these need clarification before a query generator can reliably implement the logic.

BLOCKING (completeness): Logic conditions 1 and 2 ('Add entry Toxicity Grade Date overlaps' / 'gaps') never define what constitutes an overlap or a gap between Toxicity Grade Date records. There's no stated rule for how consecutive add-entry records are ordered/paired, what tolerance (e.g., must Stop Date + 1 day = next Onset Date) defines 'no gap', or what counts as touching vs overlapping. → Add explicit definitions, e.g. 'Sort add-entry records by Tox Grade Onset Date within AE; a gap exists when the next record's Onset Date is more than 1 day after the previous record's Stop Date; an overlap exists when the next record's Onset Date is on or before the previous record's Stop Date.'

WARNING (completeness): Report Scenarios section only says 'Test Case Workbook' with no actual scenarios listed, so there's no way to verify each Logic condition (1, 2, 3, 4a, 4b, 4c, 5) maps to a distinguishable, testable scenario. → Either embed representative scenarios in the spec (input data + expected output row) for each of the 6 logic conditions, or attach/reference the actual workbook content so reviewers/generators can validate coverage.

WARNING (completeness): Column Heading includes 'Tox Grade Onset Date' and 'Tox Grade Stop Date' as single columns, but for overlap/gap and multi-record scenarios (1, 2, 3, 4a-4c) there may be multiple add-entry records involved per error row; the spec doesn't say which record's onset/stop date populates these columns when multiple are relevant (e.g., for a gap, is it the earlier record's Stop and later record's Onset, or something else). → Clarify which specific Tox Grade record's Onset/Stop values populate columns 8-9 for each scenario type (gap, overlap, AE Onset mismatch, AE End mismatch).

WARNING (ambiguity): Rule states 'Add-entry records with both dates as UNK/ND/NA are not be considered for scenarios 1 to 4 when there are other records with data entered.' It's unclear what happens when ALL add-entry records have both dates as UNK/ND/NA (no records 'with data entered') -- are they then considered, or does scenario 5 ('Add entry missing') apply instead, and how do these two rules interact for AE-level determination? → State explicitly: if all add-entry records have both dates UNK/ND/NA, treat as no add-entry present, triggering scenario 5 logic (or clarify the exact precedence between scenario 5 and scenarios 1-4 in this case).

WARNING (ambiguity): Scenario 5 says AE Term is completed but 'Add-entry section is blank... because it's not initialized or because all the records are completed with UNK/ND/NA.' This seems to overlap with the exclusion rule for scenarios 1-4, but it isn't clear whether 'all records completed with UNK/ND/NA' means both dates UNK/ND/NA (as in the exclusion rule) or just one date being UNK/ND/NA per record. → Use consistent terminology between scenario 5 and the exclusion note -- specify explicitly whether 'blank'/UNK/ND/NA refers to both onset and stop dates or either one, matching the phrasing used in the earlier exclusion rule.

INFO (ambiguity): The error messages for scenarios 1 and 2 use a literal placeholder format 'dd-mon-yyyy' without specifying which dates (from which specific records) get substituted, especially for gaps spanning two records (previous stop date and current onset date, or vice versa). → Give a concrete example, e.g. 'Overlap with Tox Grade Onset 05-Jan-2020 and Tox Stop 10-Jan-2020' showing exactly which record's dates are used, especially to disambiguate multi-record gap scenarios.

INFO (consistency): Description mentions only two triggering conditions (AE Onset ≠ earliest Tox Grade Onset OR AE End ≠ latest Tox Grade Stop), but Logic lists five distinct conditions including overlaps, gaps, and missing add-entry (scenario 5), which are not mentioned in the Description at all. → Update the Description to summarize all five logic conditions (overlap, gap, onset mismatch, end mismatch, missing add-entry) so it accurately reflects the full report scope.

WARNING (completeness): The 'Randomized only' parameter and deleted-records exclusion are stated in Logic, but there's no Report Scenario confirming behavior when the parameter is checked/unchecked or when deleted records exist, and no column indicates Randomization Number despite it being a filter criterion. → Add test scenarios covering the Randomized-only filter (checked and unchecked) and a deleted-record exclusion case; consider whether Randomization Number should be a display column for traceability.

4. Generated Code

CQL — Guardrails: 1 check(s) failed (balanced_parens)

```
SELECT `Country`, `Site`, `Subject Number`, `AE Seq`, `AE Term`, `AE Onset Date`, `AE End Date`,
`Tox Grade Onset Date`, `Tox Grade Stop Date`, `Error Message`
FROM (

-- 1) Overlap
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS Subject_Number,
    ae.AESPIDIT AS AE_Seq,
    ae.AETERMIT AS AE_Term,
    ae.AESTDTCIT AS AE_Onset_Date,
    ae.AEENDTCIT AS AE_End_Date,
    cur.AEAETOXIT AS Tox_Grade_Onset_Date,
    cur.AEAEENDTCIT AS Tox_Grade_Stop_Date,
    CONCAT('Overlap with Tox Grade Onset ', DATE_FORMAT(prev.AEAETOXIT, '%d-%b-%Y'), ' and Tox Stop
', DATE_FORMAT(prev.AEAEENDTCIT, '%d-%b-%Y')) AS Error_Message
  FROM `EDC`.`AEFM` ae
  JOIN `EDC`.`AEFM` cur ON cur.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND cur.AESPIDIT =
ae.AESPIDIT
  JOIN `EDC`.`AEFM` prev ON prev.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND prev.AESPIDIT
= ae.AESPIDIT
    AND prev.AEAETOXIT < cur.AEAETOXIT
    AND NOT EXISTS (
      SELECT 1 FROM `EDC`.`AEFM` mid
      WHERE mid.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND mid.AESPIDIT = ae.AESPIDIT
        AND (mid.DELETEDFM = 'N' OR mid.DELETEDFM IS NULL)
        AND `@Form`.`Status` = 'submitted__v'
        AND mid.AEAETOXIT > prev.AEAETOXIT AND mid.AEAETOXIT < cur.AEAETOXIT
    )
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL)
    AND (cur.DELETEDFM = 'N' OR cur.DELETEDFM IS NULL)
    AND (prev.DELETEDFM = 'N' OR prev.DELETEDFM IS NULL)
    AND NOT (prev.AEAETOXIT IS NULL AND prev.AEAEENDTCIT IS NULL)
    AND NOT (cur.AEAETOXIT IS NULL AND cur.AEAEENDTCIT IS NULL)
    AND prev.AEAEENDTCIT IS NOT NULL
    AND cur.AEAETOXIT <= prev.AEAEENDTCIT
    AND (NOT @RandomizedOnly OR ae.AE_O_SAFRAND_DRV IS NOT NULL)

  UNION ALL

-- 2) Gap
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS Subject_Number,
    ae.AESPIDIT AS AE_Seq,
    ae.AETERMIT AS AE_Term,
    ae.AESTDTCIT AS AE_Onset_Date,
    ae.AEENDTCIT AS AE_End_Date,
    cur.AEAETOXIT AS Tox_Grade_Onset_Date,
    cur.AEAEENDTCIT AS Tox_Grade_Stop_Date,
    CONCAT('Gap with previous Tox Grade Onset ', DATE_FORMAT(prev.AEAETOXIT, '%d-%b-%Y'), ' and Tox
Stop ', DATE_FORMAT(prev.AEAEENDTCIT, '%d-%b-%Y')) AS Error_Message
  FROM `EDC`.`AEFM` ae
  JOIN `EDC`.`AEFM` cur ON cur.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND cur.AESPIDIT =
ae.AESPIDIT
  JOIN `EDC`.`AEFM` prev ON prev.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND prev.AESPIDIT
= ae.AESPIDIT
    AND prev.AEAETOXIT < cur.AEAETOXIT
    AND NOT EXISTS (
      SELECT 1 FROM `EDC`.`AEFM` mid
      WHERE mid.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID` AND mid.AESPIDIT = ae.AESPIDIT
        AND (mid.DELETEDFM = 'N' OR mid.DELETEDFM IS NULL)
        AND `@Form`.`Status` = 'submitted__v'
        AND mid.AEAETOXIT > prev.AEAETOXIT AND mid.AEAETOXIT < cur.AEAETOXIT
    )
  WHERE `@Form`.`Status` = 'submitted__v'
```

```
    AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL)
    AND (cur.DELETEDFM = 'N' OR cur.DELETEDFM IS NULL)
    AND (prev.DELETEDFM = 'N' OR prev.DELETEDFM IS NULL)
    AND NOT (prev.AEAETOXIT IS NULL AND prev.AEAEENDTCIT IS NULL)
    AND NOT (cur.AEAETOXIT IS NULL AND cur.AEAEENDTCIT IS NULL)
    AND prev.AEAEENDTCIT IS NOT NULL
    AND cur.AEAETOXIT > SUBDATE(prev.AEAEENDTCIT, INTERVAL -1 DAY)
    AND (NOT @RandomizedOnly OR ae.AE_O_SAFRAND_DRV IS NOT NULL)

  UNION ALL

-- 3) AE Onset not earliest Tox Grade Onset
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS Subject_Number,
    ae.AESPIDIT AS AE_Seq,
    ae.AETERMIT AS AE_Term,
    ae.AESTDTCIT AS AE_Onset_Date,
    ae.AEENDTCIT AS AE_End_Date,
    tg.MinOnset AS Tox_Grade_Onset_Date,
    tg.MaxStop AS Tox_Grade_Stop_Date,
    'AE Onset not earliest Tox Grade Onset' AS Error_Message
  FROM `EDC`.`AEFM` ae
  JOIN (
    SELECT `@HDR`.`Subject`.`ID` AS SubjID, AESPIDIT AS AESeqKey,
           MIN(AEAETOXIT) AS MinOnset, MAX(AEAEENDTCIT) AS MaxStop
    FROM `EDC`.`AEFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
      AND NOT (AEAETOXIT IS NULL AND AEAEENDTCIT IS NULL)
    GROUP BY `@HDR`.`Subject`.`ID`, AESPIDIT
  ) tg ON tg.SubjID = ae.`@HDR`.`Subject`.`ID` AND tg.AESeqKey = ae.AESPIDIT
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL)
    AND ae.AESTDTCIT <> tg.MinOnset
    AND (NOT @RandomizedOnly OR ae.AE_O_SAFRAND_DRV IS NOT NULL)

  UNION ALL

-- 4a/4b/4c) AE End not latest Tox Grade Stop
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS Subject_Number,
    ae.AESPIDIT AS AE_Seq,
    ae.AETERMIT AS AE_Term,
    ae.AESTDTCIT AS AE_Onset_Date,
    ae.AEENDTCIT AS AE_End_Date,
    tg.MinOnset AS Tox_Grade_Onset_Date,
    tg.MaxStop AS Tox_Grade_Stop_Date,
    'AE End not latest Tox Grade Stop' AS Error_Message
  FROM `EDC`.`AEFM` ae
  JOIN (
    SELECT `@HDR`.`Subject`.`ID` AS SubjID, AESPIDIT AS AESeqKey,
           MIN(AEAETOXIT) AS MinOnset, MAX(AEAEENDTCIT) AS MaxStop,
           SUM(CASE WHEN AEAEENDTCIT IS NULL THEN 1 ELSE 0 END) AS BlankStopCount
    FROM `EDC`.`AEFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
      AND NOT (AEAETOXIT IS NULL AND AEAEENDTCIT IS NULL)
    GROUP BY `@HDR`.`Subject`.`ID`, AESPIDIT
  ) tg ON tg.SubjID = ae.`@HDR`.`Subject`.`ID` AND tg.AESeqKey = ae.AESPIDIT
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL)
    AND (
      (ae.AEENDTCIT IS NOT NULL AND (tg.BlankStopCount > 0 OR ae.AEENDTCIT <> tg.MaxStop))
      OR
      (ae.AEENDTCIT IS NULL AND tg.BlankStopCount = 0)
    )
```

```
    AND (NOT @RandomizedOnly OR ae.AE_O_SAFRAND_DRV IS NOT NULL)

  UNION ALL

-- 5) Add entry missing
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS Subject_Number,
    ae.AESPIDIT AS AE_Seq,
    ae.AETERMIT AS AE_Term,
    ae.AESTDTCIT AS AE_Onset_Date,
    ae.AEENDTCIT AS AE_End_Date,
    CAST(NULL AS DATE) AS Tox_Grade_Onset_Date,
    CAST(NULL AS DATE) AS Tox_Grade_Stop_Date,
    'Add entry missing' AS Error_Message
  FROM `EDC`.`AEFM` ae
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL)
    AND ae.AETERMIT IS NOT NULL
    AND NOT EXISTS (
      SELECT 1 FROM `EDC`.`AEFM` tg
      WHERE tg.`@HDR`.`Subject`.`ID` = ae.`@HDR`.`Subject`.`ID`
        AND tg.AESPIDIT = ae.AESPIDIT
        AND `@Form`.`Status` = 'submitted__v'
        AND (tg.DELETEDFM = 'N' OR tg.DELETEDFM IS NULL)
        AND NOT (tg.AEAETOXIT IS NULL AND tg.AEAEENDTCIT IS NULL)
    )
    AND (NOT @RandomizedOnly OR ae.AE_O_SAFRAND_DRV IS NOT NULL)
) results
ORDER BY Country, Site, Subject_Number, `AE Seq`
```

I mapped the AE domain to Form AEFM (from the confirmed metadata list) using AESPIDIT as the AE sequence number, AETERMIT as the AE term, AESTDTCIT/AEENDTCIT as AE onset/end dates, and AEAETOXIT/AEAEENDTCIT as the toxicity-grade add-entry onset/stop dates within the same AEFM record (assumption: since no separate AGTOX-style child Form is listed in the study metadata, the toxicity grade onset/stop dates are modeled as repeating items AEAETOXIT/AEAEENDTCIT within AEFM itself -- this should be confirmed via DESCRIBE FORM against the real add-entry structure, as it may actually be a separate repeating section not captured in the flat item list given). Each of the five error scenarios (overlap, gap, AE-onset-mismatch, AE-end-mismatch covering 4a/4b/4c, and missing add-entry) is expressed as a UNION ALL branch filtered to submitted/non-deleted records, excluding "both dates UNK/ND/NA" add-entry rows via the NOT(...IS NULL AND ...IS NULL) checks, and gated by the @RandomizedOnly parameter checked against AE_O_SAFRAND_DRV (assumed to hold the randomization-number derived flag, since no dedicated RANDOM form exists in metadata -- flagged as an assumption). Country is pulled from `@HDR`.`Site`.`Country` per the required convention rather than a CountryName property.